



Networks and Systems: MM3

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- **Mm1** Introduction to Systems and Networks (hps)
- **Mm2** Layer-2 Example Technologies: Field-buses CAN and Flexray (hps)
- **Mm3** Clock Synchronisation & impact of clock offsets on sensor data (hps)
- **Mm4** Network Calculus I (HS)
- **Mm5** Network Calculus II (HS)
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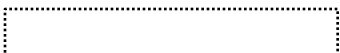


MM3 Learnings

- Understand/Revise how clocks and clock synchronization in distributed systems work
- Understand how sensor data in some cases can be used to identify clock offsets
- Understand the impact of clock offsets on distributed measurements
- Get some insights into the systems-of-systems use-case of monitoring & control of electricity grids

plus - on the methodology side:

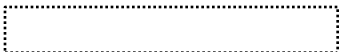
- Revise your knowledge of continuous time Markov chains
- Understand how you can use/work with continuous time Markov chains to model time series of sensor data





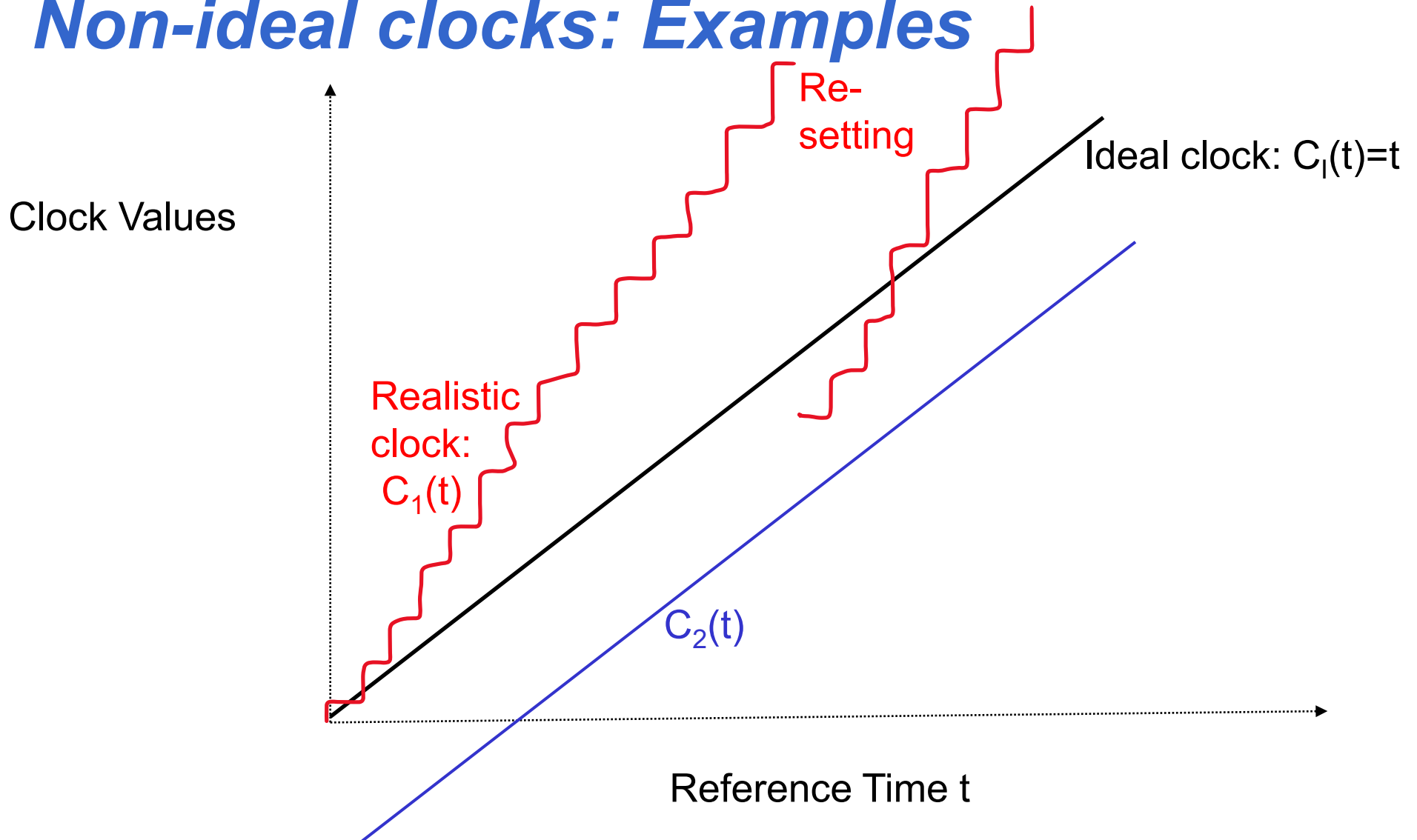
Intro: Distributed System

- Set of N Nodes
 - Interacting via a communication network
 - Collaborating on an overall task
 - Illustration in space-time diagrams (message sequence charts)
- Challenges
 - Non-ideal communication networks (see earlier lectures)
 - Varying (stochastic) delays
 - Message losses - recoverable by retransmission schemes
 - Message reorderings - pairwise FIFO order can be achieved via sequence numbers
 - **Use of time from local clocks**
 - (Accidental) faults and malicious behavior (see earlier lectures)





Non-ideal clocks: Examples



Intro: Time and clocks

- Distributed system with N nodes
- Local clocks $C_k(t)$ at component k at absolute 'reference' time t
 - monotonicity
 - granularity $g \rightarrow C_k(t)$ not continuous
 - when assuming continuous, differentiable $C_k(t)$ [possibly with 'resetting' jumps]
 - drift: $\rho_k > 0$ such that $|dC_k(t)/dt - 1| < \rho_k$
- in order to make local time-stamps in the distributed system useful
 - bounded precision: $|C_i(t) - C_k(t)| \leq \pi$
 - bounded accuracy: $|C_i(t) - t| \leq \alpha$
- Methods
 - clock synchronisation protocols
 - use of 'logical' clocks and 'ordering'
 - algorithms that do not rely on time or order of distributed events



Different notions of time

- Coordinated Universal Time (UTC)
 - International time based on atomic clocks
 - occasionally adjusted to astronomic time (leap seconds, etc.)
 - broadcasted via radio stations and satellite (GPS)
 - accuracy:
 - land-based signals: 0.1-10ms
 - GPS: $<1\mu\text{s}$
- Internal clocks
 - based on counters $H_i(t)$ with some increment frequency
 - for generation of time-stamps: $C_i(t)=a_i*H_i(t)+b_i$
 - ordinary quartz clocks: $\sim 1\text{sec}$ drift in 11-12 days
- use of time-stamps, e.g.
 - Timestamping sensor data
 - event ordering
 - life-times of authentication tickets / avoiding replay attacks



Content

1. Introduction

2. Clock Synchronisation protocols

- Basic methods for synchronization and properties
- Network Time Protocol (ntp)

3. System-of-systems use-case: digital twin of electricity distribution grids

- Background MV and LV grids
- Measurements
- Digital twin
- Offset detection in distributed voltage measurements

4. Impact of clock offset on measurements

- Use-case loss calculation
- definition of time alignment error
- Modelling of consumption measurements via Markov modulated processes

5. Summary

Clock synchronisation I

- Scenario:
 - Distributed system with N components
 - Local clocks $C_k(t)$ at each component k
- Goal: Define protocol such that clock precision immediately after execution is minimized
 - $|C_i(t_0) - C_k(t_0)| \leq \pi$
 - possibly also adjust drift
- Types
 - internal synchronisation
 - external $\rightarrow$ authoritative time source
- First approach (client-server based)
 - Client A contacts Server B
 - B sends its current time-stamp $t_B = C_B(t)$ to A
 - A knows bounds min, max on the one-way communication delay
 - A sets its clock to $t_B + (\max - \min)/2$

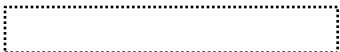
Clock synchronisation II

- Christian's method
 - Client A contacts Server B (request, time-stamps for send and receive)
 - B sends its current time-stamp $t_B = C_B(t)$ to A
 - A estimates the round-trip time $\check{R}$
 - A sets its clock to $t_B + \check{R}/2$
 - Precision = $\check{R}/2$
- Inaccuracies and other problems
 - asymmetric communication links
 - processing delays
 - delay variations (jitter)
 - $\text{drift}(\text{clock A})$ deviates from $\text{drift}(\text{clock B})$
 - Clock monotonicity
 - Time server may fail/be faulty



Clock synchronisation III

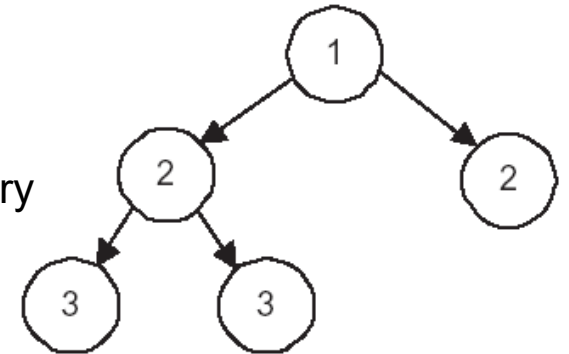
- Berkeley algorithm: internal synchronisation of group of computers
 - Master polls clock values from slaves
 - uses round-trip time estimates to obtain slaves' clock values
 - elimination of out-liers (large RTTs, strong deviations from average)
 - form average and send required adjustments to slaves
 - upon failure of master → re-election by group



Clock synchronisation IV: NTP

Network Time Protocol (NTP)

- Version 4 defined in RFC5905 (see <https://tools.ietf.org/html/rfc5905>)
- architecture for time-service and (UDP-based) protocols to distribute time info over the Internet
- Goal: synchronisation of clients to UTC
- Hierarchical organisation of servers (strata)
 - primary servers connected to UTC source
- Reconfiguration of servers upon failures
 - primary server with loss of UTC connection becomes secondary
 - re-selection of lower-stratum server upon failure
- Synchronisation Modes
 - Multicast (within LAN, using delay assumptions)
 - Client-Server
 - Symmetric (peer-2-peer)





NTP (cntd)

NTP timestamps

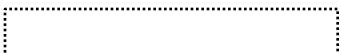
- 64 bit unsigned fixed-point number (32 bit integer part)
- Representing seconds since 0h, Jan. 1, 1900

Messages:

- All messages are time-stamped
- Each message contains time-stamps of recent events
 - Local time of send & receive of previous message
 - Local time of send of current message

Functionality:

- Determine offset o and delay d for each message pair
- Filtering of $\langle o, d \rangle$ time series for improved accuracy
- Variance estimates for peer selection
- Accuracy: LAN ~1ms, Internet ~tens of ms



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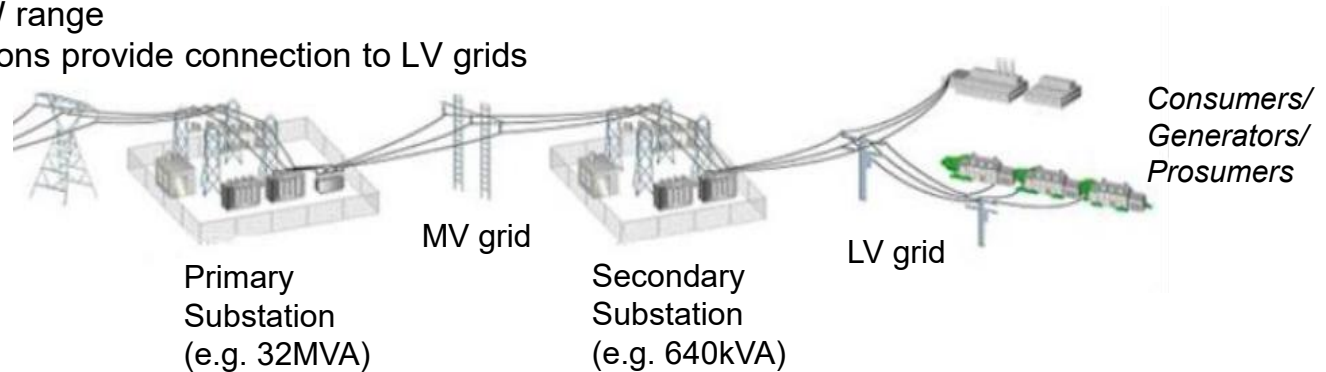
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Electricity Distribution Grids

Hierarchical electricity grid structure

- Medium Voltage (10kV, 20kV or 30kV)
 - Consumers & generators ~ MW range
 - ~50...~500 Secondary substations provide connection to LV grids
- Low Voltage (400V)
 - Consumers & generators ~ kW range
 - 1...~120 customer connections per secondary substation



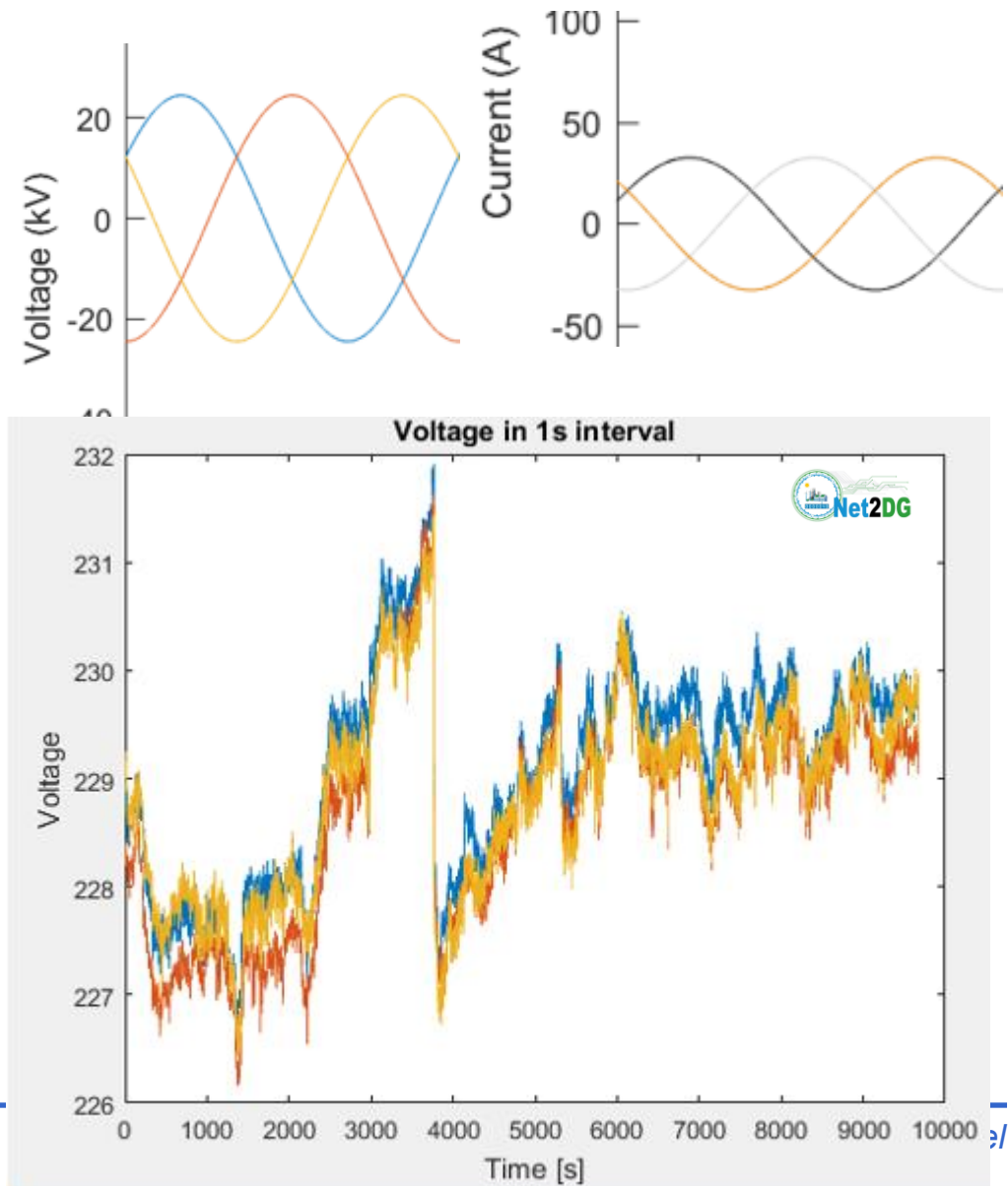
Targets of Distribution Grid Operator

- Reliable distribution of electrical energy
 - minimize outage durations & impact, avoid overload and faults
- Voltage quality
 - Assure voltage within $\pm 10\%$ (or $\pm 6\%$) of nominal voltage
- Efficiency
 - Minimize energy losses in the grid
 - Increase life-time of grid components
 - Target grid investments for renewable generation integration and new consumption

Physical System: Electricity Distribution Grids – 3-Phase AC Grids

Alternating Current (AC)

- Physical parameters:
Currents $I(t)$ and voltages $V(t)$ at each grid location for each phase
 - e.g. 50 Hz, 20ms cycle
- Sub-cycle behavior
 - Differential algebraic equations
 - EMT models (Electromagnetic Transient)
 - time-steps of few μs
- RMS values - Averages over cycles
 - represent phase angles by complex voltages/currents (phasors)
 - Power flow equations:
non-linear equation system
 - Often applied at time-resolution of minutes to hours





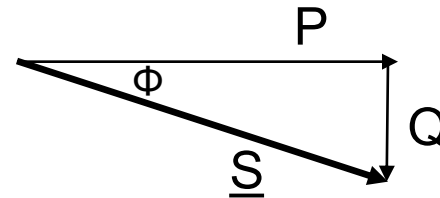
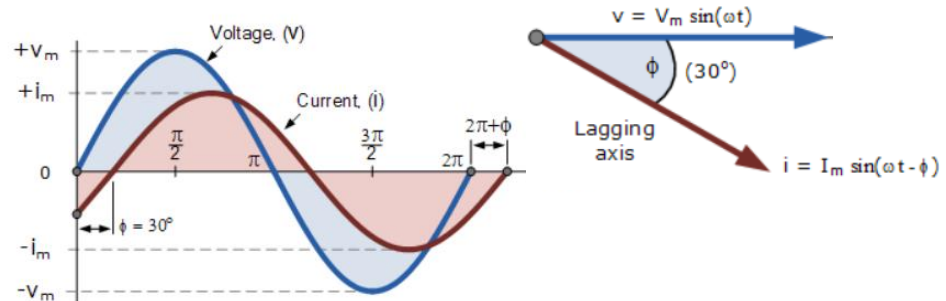
Electrical Parameters for AC grids - Basics

Base Parameters

- Voltage Magnitude (Volt)
 - and phase angle (compared to some reference point in the grid)
→ Voltage phasor: $\underline{U}$
- Current Magnitude (Amper)
- Phase shift (angle) between voltage and current
 - Power factor: $\cos(\phi)$

Power: $\underline{S} = \underline{U} \underline{I}^* = P + j Q$

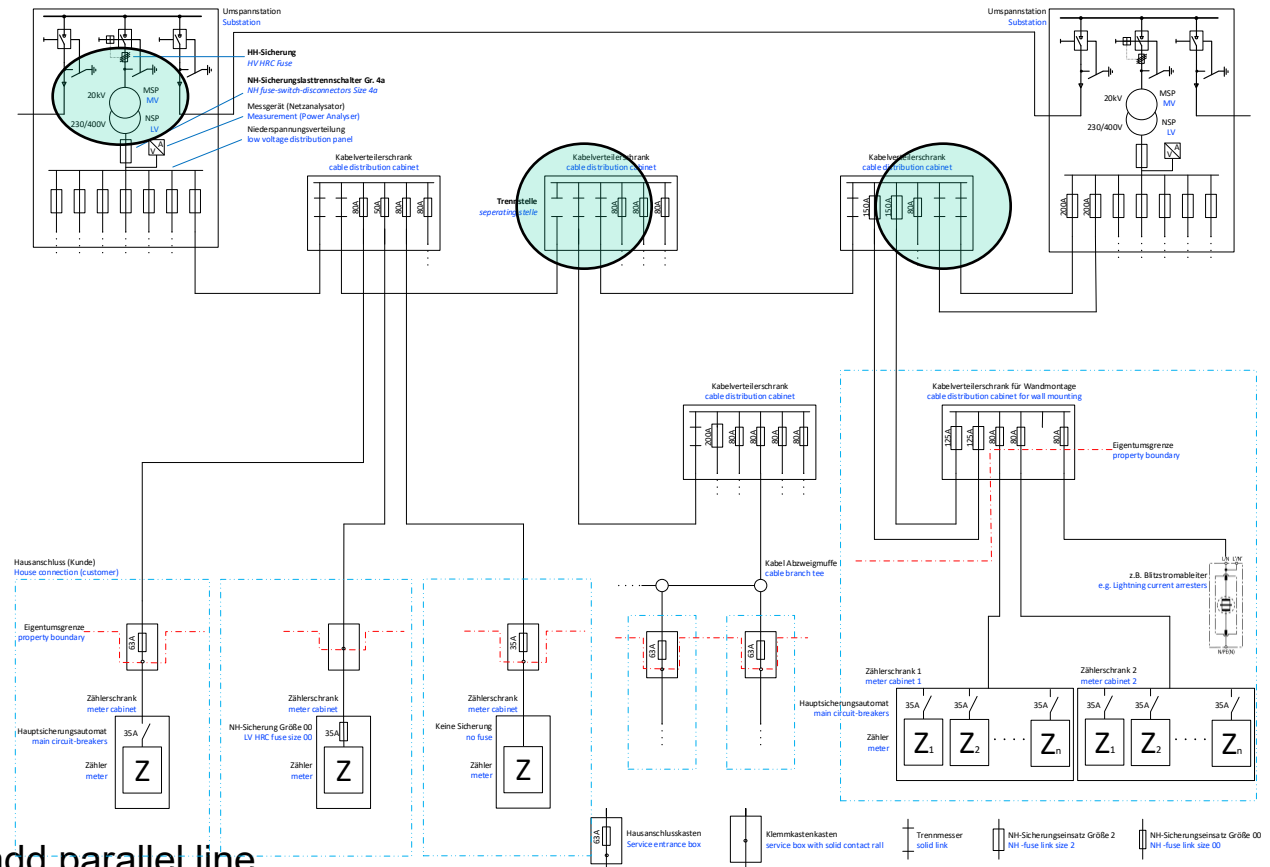
- P active power – unit kW
- Q reactive power – unit kVar
- $S = |\underline{S}|$ apparent power – unit kVA
 - Power factor = $\cos \Phi = P/S$
 - Same Φ as between $\underline{U}$ and $\underline{I}$



- Active resp. Reactive Energy = Integral of P resp. Q over time

Grid optimizations - possibilities

- **Trafo setting**
 - Changes ratio primary – secondary
(e.g. 20000:400, 20800:400, ...)
 - Voltage management
- **Topology adjustments**
 - Open ,ring structure
 - Partially remote switching in MV grids otherwise manual
- **Replace with larger grid components**
 - Trafo size
 - Line diameter, line material; add parallel line
- **Reactive power management of consumers or generators or within the grid**
- **Active power management:** demand response, generation curtailment



Data sources: Smart Meters

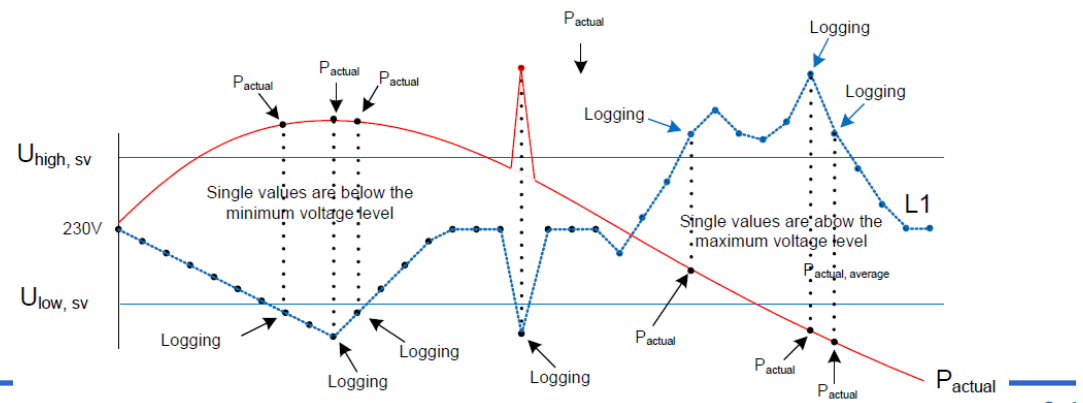
Measuring

- Electrical consumption
- Voltage levels
- Current
- Power factor
- Harmonic distortions
- Phase angles

e.g., 1 second measurement data internally
15min or even 1hour values (avg, min, max)
remotely sent

Access via

- Periodic uploads
- Alarms
- On-Demand Data Access



Electrical Measurement Devices

Voltages per phase, currents, power, power-factor, harmonics ...

- Before the busbar at the secondary substation
 - On each LV feeder (power/currents)
 - Optionally also in junction boxes
- Variant: voltage msrt. in PLC Repeater

Resolution ~5s...15min

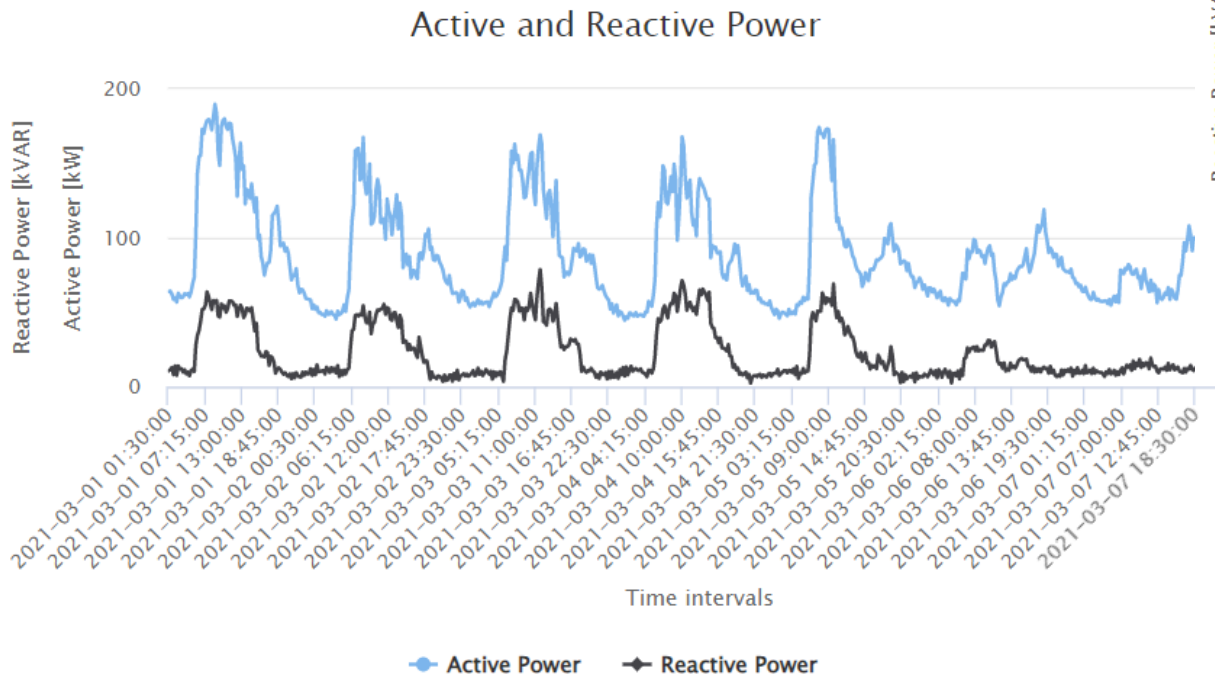
Readout of values over local interface (Modbus)

- Remotely connected over
e.g. fiber,
2-5G, ...



Secondary substation

- Example 630kVA station in DK (with little PV)
7 days early March



Convention (here and frequent):

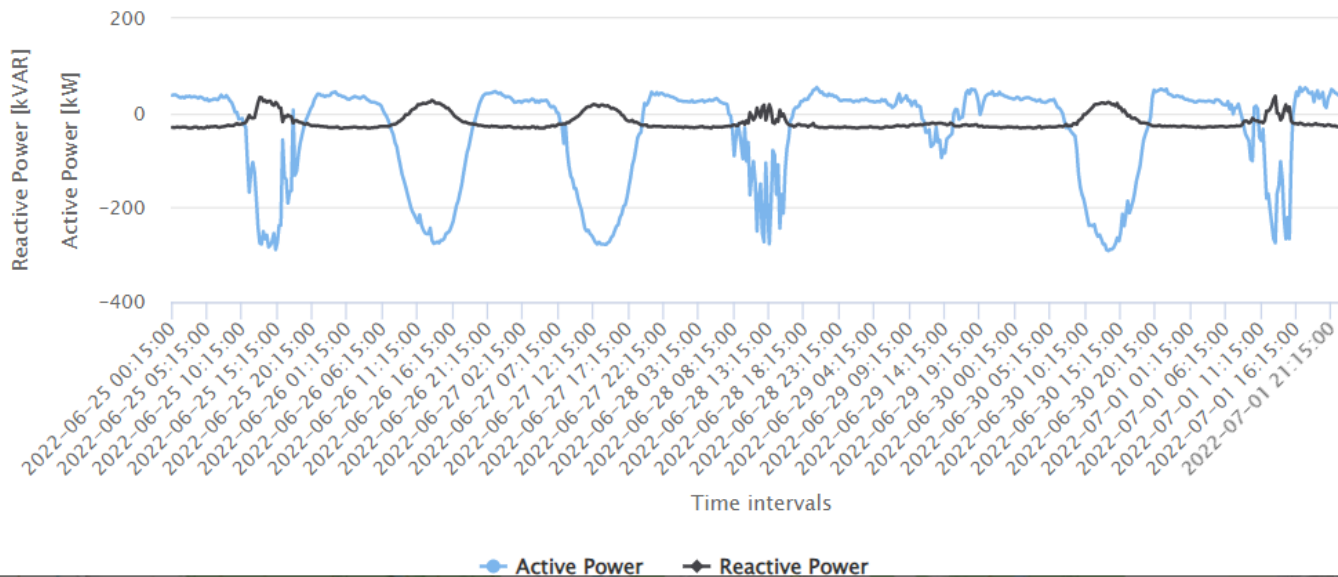
positive active power = consumption, negative active power = generation;

positive reactive power = inductive behavior, negative reactive power = capacitive behavior;

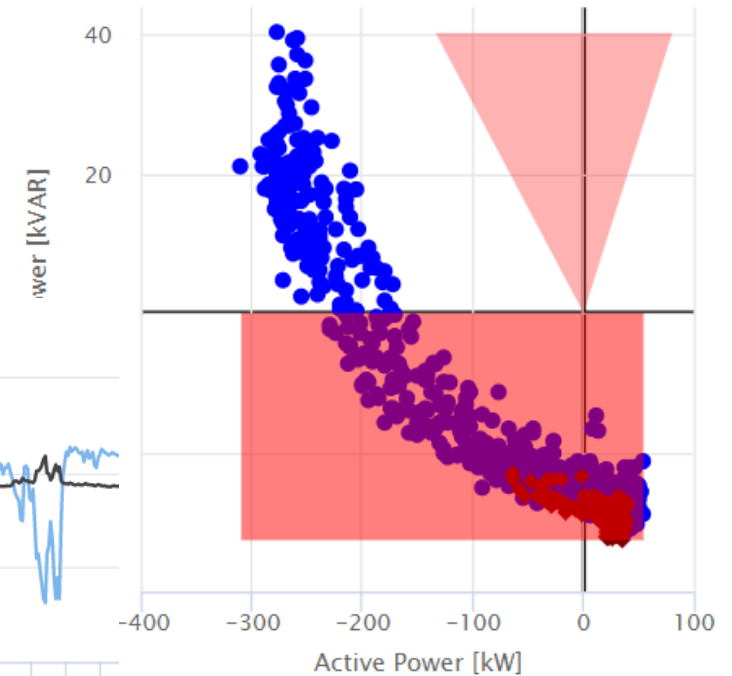
Secondary substation

- Example 400kVA station with PV in Germany
7 days end of June

Active and Reactive Power

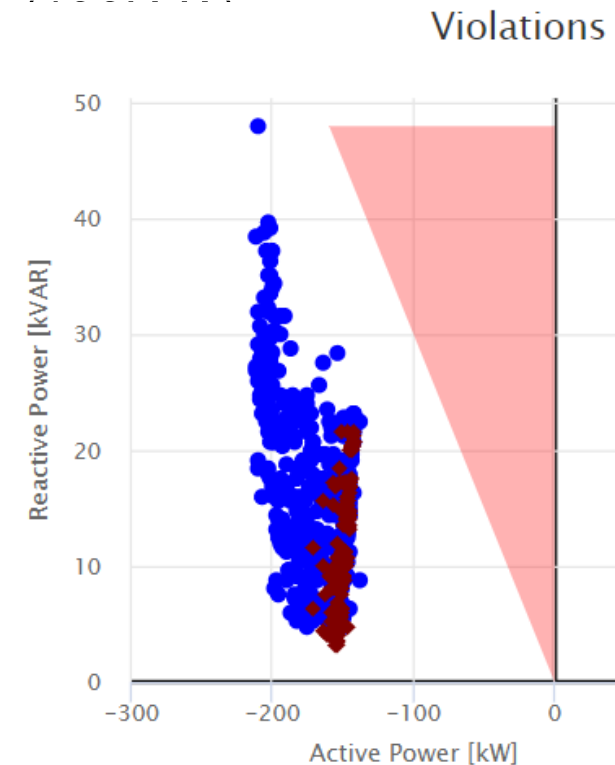
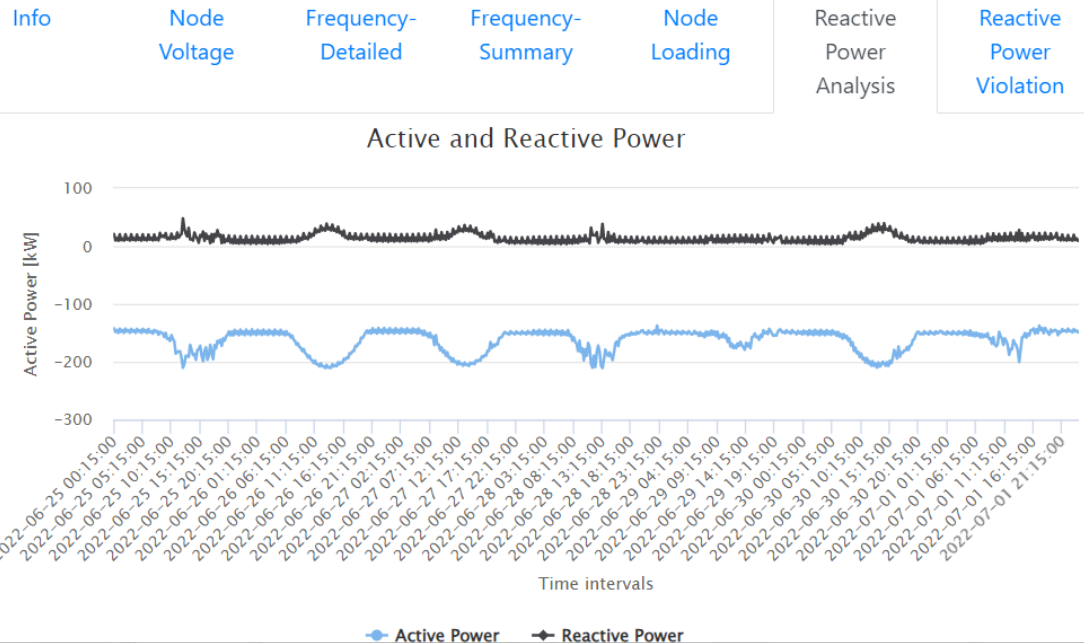


Violations



Prosumer Measurement – different types

- Large customers at own secondary substation



Measurement errors – value errors

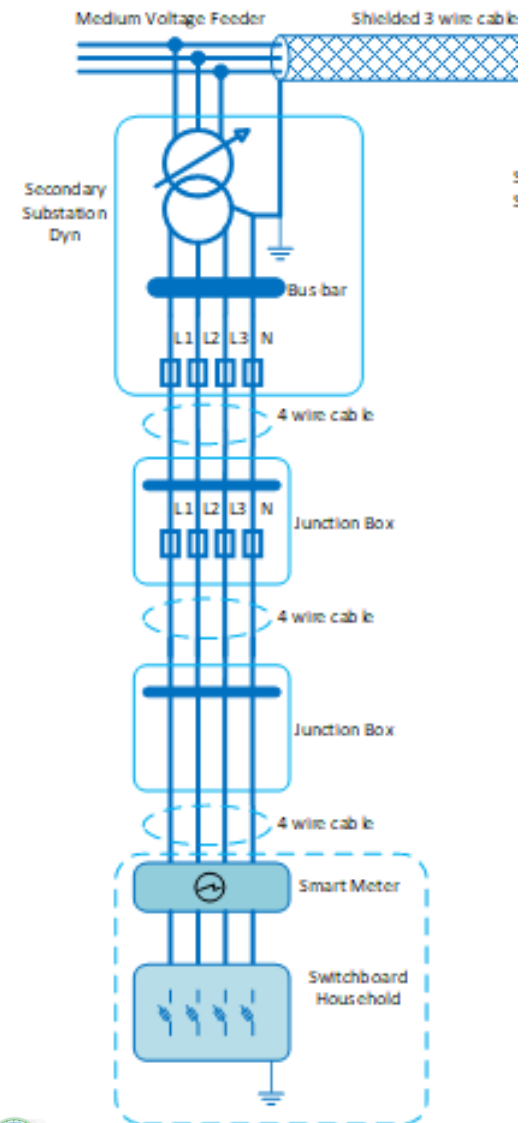
- Depend on device class and on parameter measured
e.g. 0.1% ... 1%
 - Percentage can refer to the maximally measurable value
- Not only the measurement device class matters, but also
 - Voltage and Current transformers used for connecting the device

Art kind	Klasse class	Strommessabweichung in % bei ... current error in % at ...							
		1	5	20	50	100	120	150	200
		Prozent der Bemessungsstromstärke (I_N) percent of rated current							
1	0.1		0.4	0.2		0.1	0.1		
	0.2		0.75	0.35		0.2	0.2		
	0.2 S	0.75	0.35	0.2		0.2	0.2		
	0.5		1.5	0.75		0.5	0.5		
	0.5 S	1.5	0.75	0.5		0.5	0.5		
	1		3	1.5		1	1		
	3				3		3		
	5				5		5		



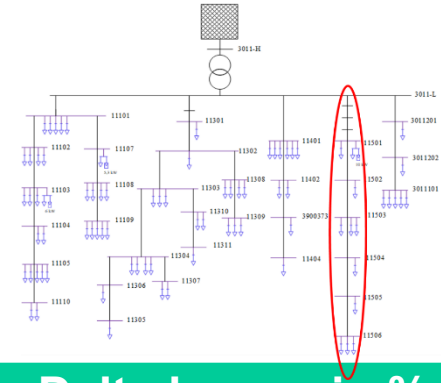
Target: Grid Efficiency

- Europe: on average more than 7% energy lost over local distribution systems
- Technical Losses
 - Variable: lines, cables, copper parts of transformers
 - proportional to square of current, I^2
 - Fixed: transformer cores
 - Can be influenced by
 - Transformer dimensioning
 - Cable resistance reduction
 - Grid topology changes
 - Phase balancing
 - Demand and generation balancing
- Non-technical Losses: non-measured (not billed) consumption
 - Malicious causes: theft
 - Accidental causes, e.g. faulty meters
 - Consumption by DSO operating staff



Reduction of technical losses: TME feeder example

- Loss reduction possible through
 - Changes in the topology
 - **Voltage management**
 - **Reactive power management**



Technical losses in feeder within 24 h	Base voltage	Delta losses in %
1.05 kWh	380 V	+12.0 %
0.93 kWh	400 V	<i>Base Case</i>
0.83 kWh	424 V	-11.6%
Total Power loss	PV Inverters	
0.95 kWh	Standard Q(U) local control	<i>Base Case</i>
0.72 kWh	React. Power Minimization @SS	-19.8%



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Joint processing of measurements in distribution grids

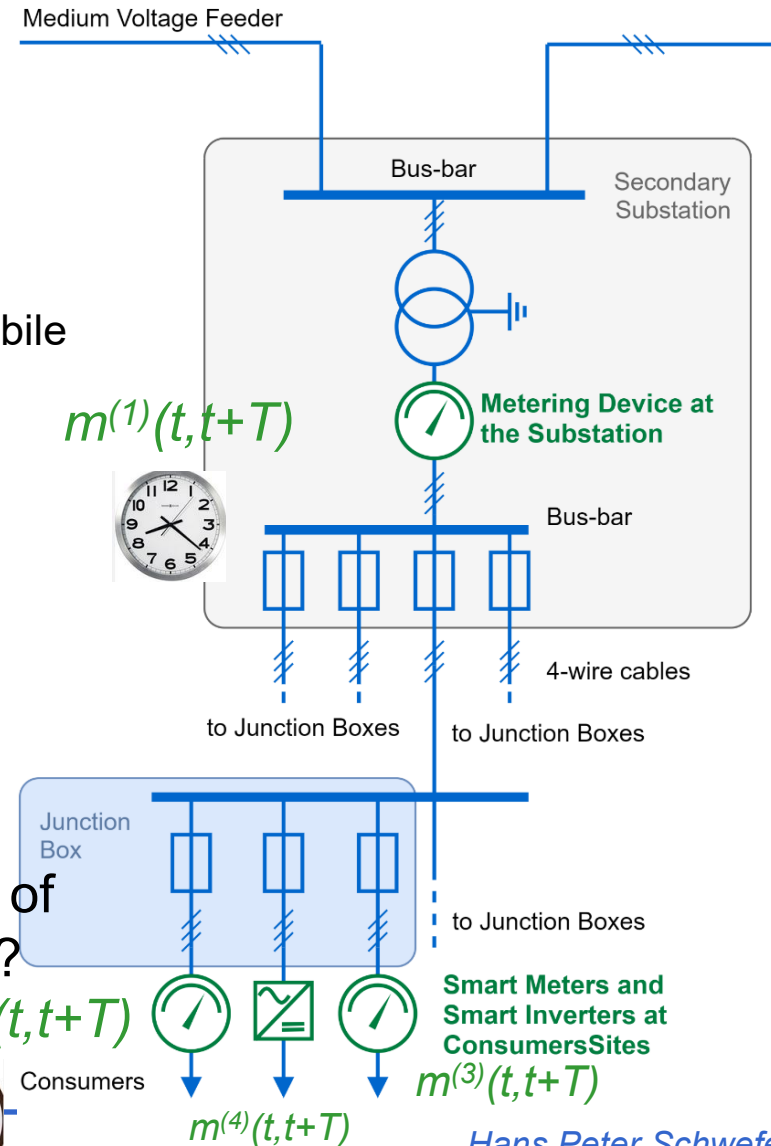
Heterogeneous sensors

- Customer-site: inverters, Smart Meters
- Power (Quality) measurement devices -stationary or mobile
- Etc.

Need to correlate measurements from different grid locations, e.g. for

- Fault detection & diagnosis (voltages, currents)
- Energy loss calculations (power, energy)

→ What is the impact of non-ideal clocks?





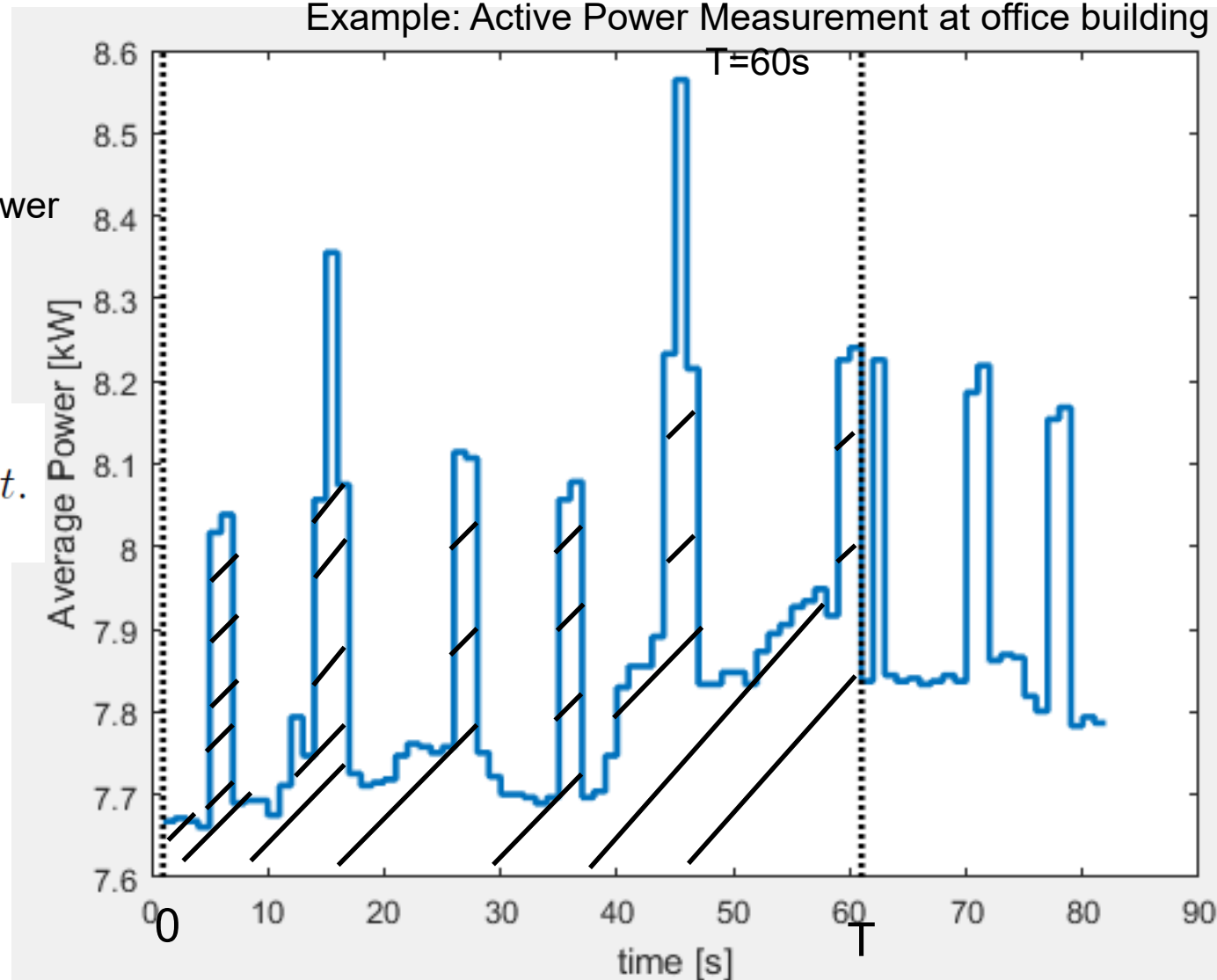
Single measurement point:

- Measurand $m(t)$, e.g. active power
- Measurement device provides average over interval of size T :

$$\hat{m}(T) := 1/T \int_0^T m(t) dt.$$

→ What 'error' will be introduced by a clock offset δ ?

Example: Active Power Measurement at office building





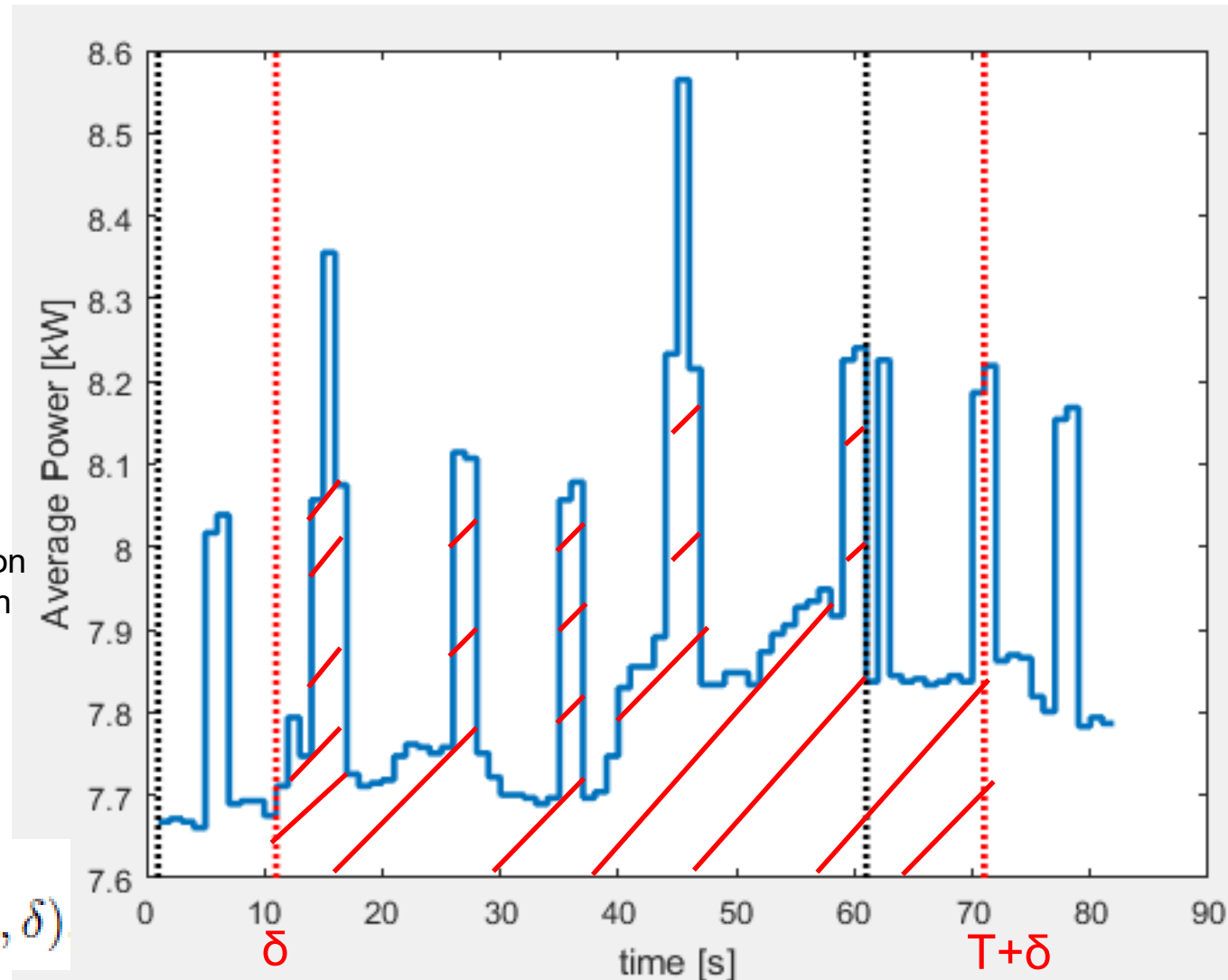
- Clock deviations lead to different measurement intervals, e.g. for clock offset δ : $[\delta, T + \delta]$

$$\hat{m}(T, \delta) := 1/T \int_{\delta}^{T+\delta} m(t) dt.$$

- Clock offset δ
 - depends on synchronization protocols & communication network

- Deviation in value caused by clock offset:

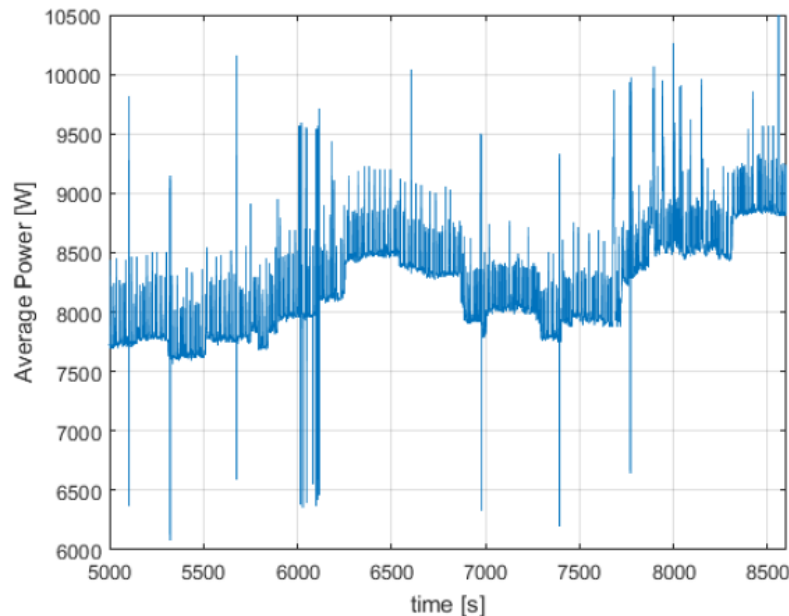
$$\epsilon(T, \delta) := \hat{m}(T, 0) - \hat{m}(T, \delta)$$



Empiric distribution of measurement deviation

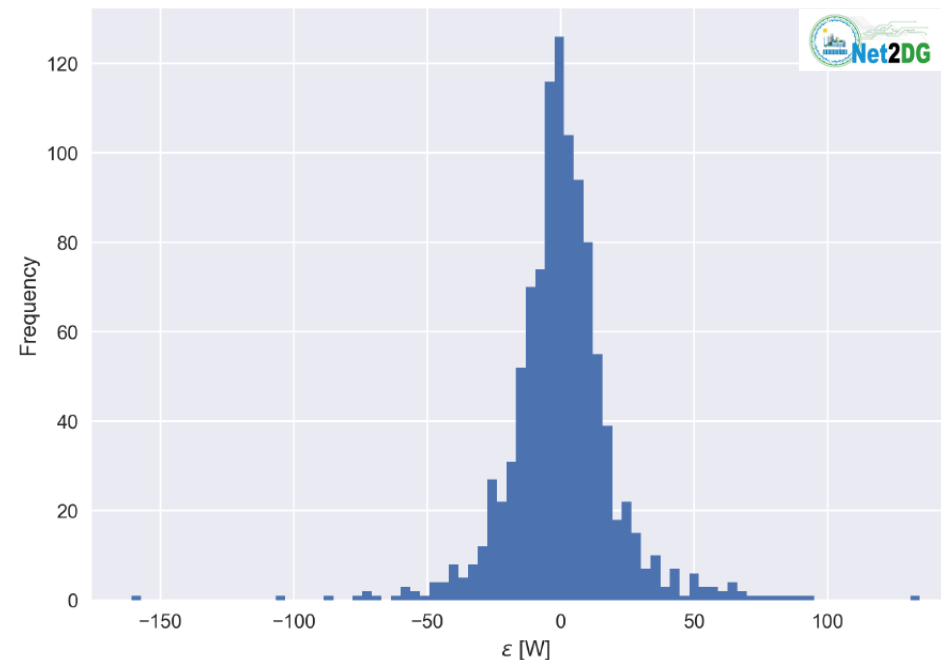
$$\hat{m}(T, \delta) := 1/T \int_{\delta}^{T+\delta} m(t) dt. \quad \epsilon(T, \delta) := \hat{m}(T, 0) - \hat{m}(T, \delta),$$

Measurement Deviation: map offset delta in time domain of averaging interval to deviation of resulting



$m(t)$: power measurement at office building (1sec intervals)

Trace: $T=100$, $\delta=8$, mean= $4.1649\text{e-}01$, std= $2.0649\text{e+}01$

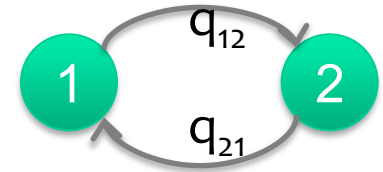


Distribution of ϵ , $T=100\text{s}$, $\delta=8\text{ s}$
 $\rightarrow$ Standard deviation $\approx 0.25\%$ of mean
 (normalized error)

Continuous time Markov Chains

Definitions

- State-space: finite or countable infinite
E.g., $E=\{1,2,\dots,N\}$ ($N=\infty$ also allowed)
- Future behavior only depends on current state
- Transition rate matrix: Q is $n \times n$ (n is number of states)
 - q_{ij} is transition rate (events/time) between states i and j , where $q_{ij} > 0$.
 - Q 's diagonal entries are negative so that rates sum to zero:
 - For homogeneous MC: $Q(t) = Q$.



$$Q = \begin{bmatrix} -q_{12} & q_{12} \\ q_{21} & -q_{21} \end{bmatrix}$$

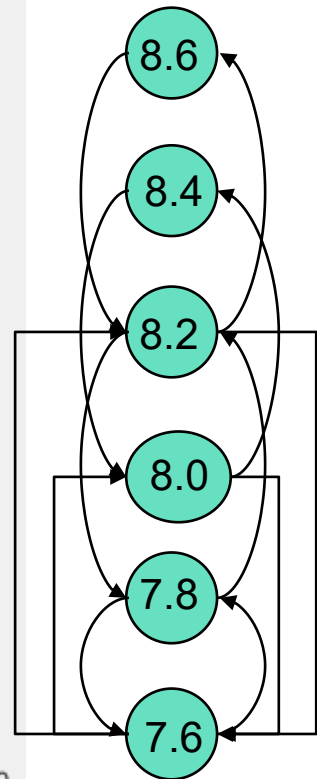
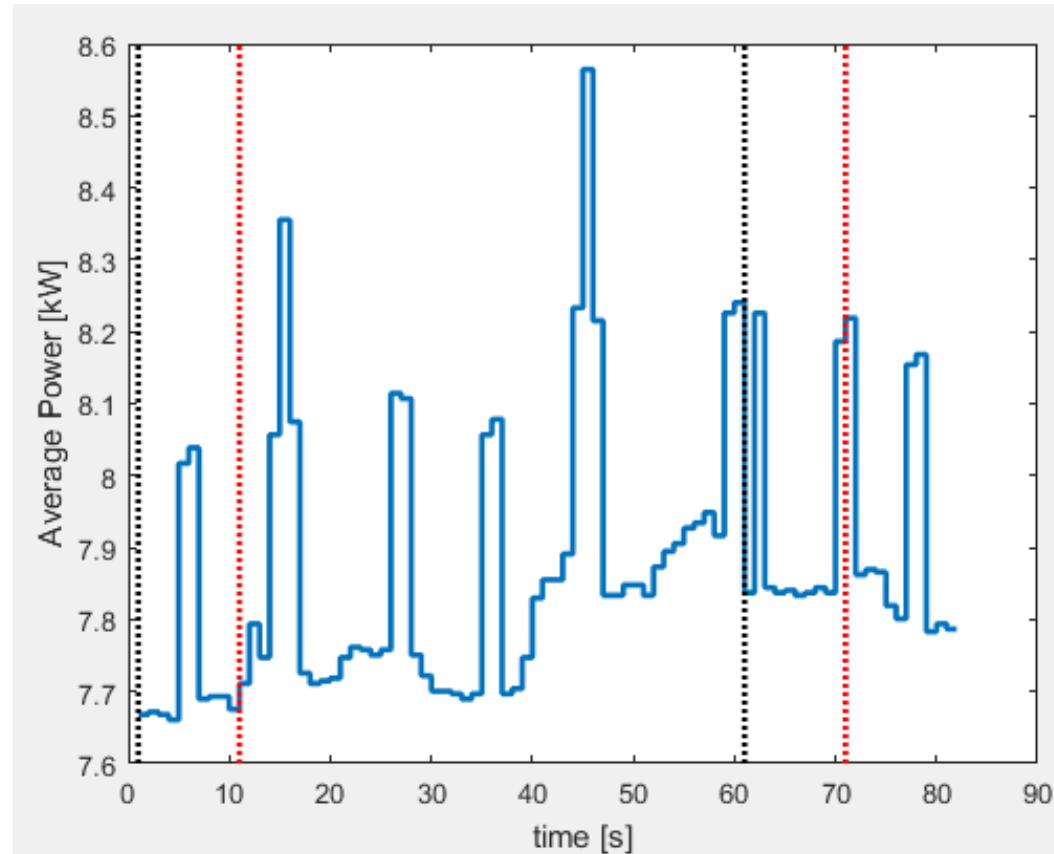
$$\sum_{all j} q_{ij} = 0$$

Properties

- State holding time (or inter-event time): D_i is exponentially distributed
 - cdf of state holding time for state i : $P[D_i \leq t] = 1 - e^{-\Lambda_i t}$, $t \geq 0$.
 - Λ_i is rate of exponential state holding time distribution: $\Lambda_i = -Q_{ii}$
- Computation of state probabilities:
 - State probabilities at time t : $P(t) = P(0) \exp(Qt)$, ($P(0)$ is initial state probabilities)
The vector $P(t)$ express likelihoods of being in each state after t time, given started with $P(0)$
 - Steady state solution: as $t \rightarrow \infty$, we expect $d\pi/dt \rightarrow 0$ (for irreducible and positive recurrent MC)
(the change in $P(t)$) should settle for large values of t
 $\pi Q = 0$ and $\pi(1, \dots, 1)' = 1$

Markov modulated process (MMP):

- Generator Matrix Q
- Power-values in $\text{diag}(E)$
 - E_{ii} = power consumed while in State i





Given continuous time trace, e.g. by $\langle t_i, v_i \rangle$

- t_i is time instant at which value changes to v_i

Fitting a Markov modulated chain with N states to the trace as follows

- 1) Discrete v_i to set of values $d_1, \dots, d_N$
- 2) Determine probability matrix P of discrete time Markov chain at change moments

$$p_{ij} = \frac{\text{\#times trace changes from value } v_i \text{ to } v_j}{\text{\#times trace changes from value } v_i \text{ to any other value}}$$

- 3) Determine total state leaving rate of continuous time Markov chain

$$\text{Lambda}_i = 1/\text{mean}(\text{state holding times of state } i)$$

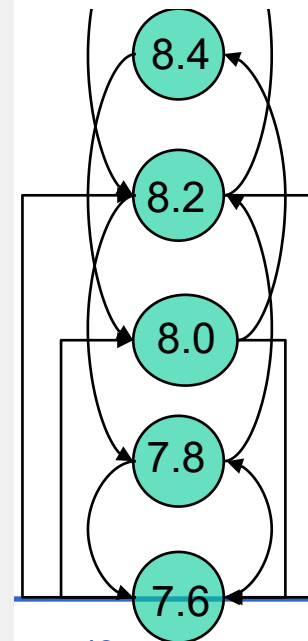
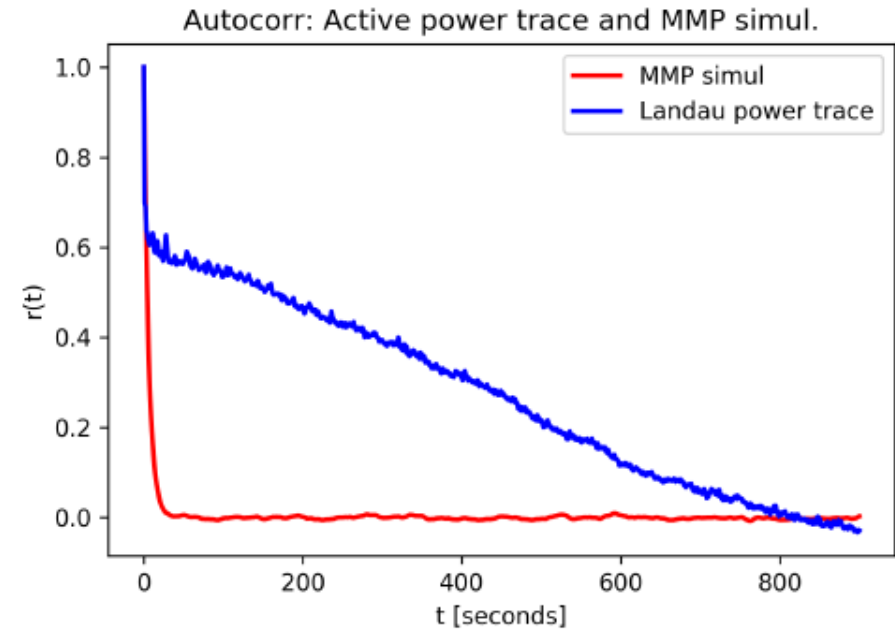
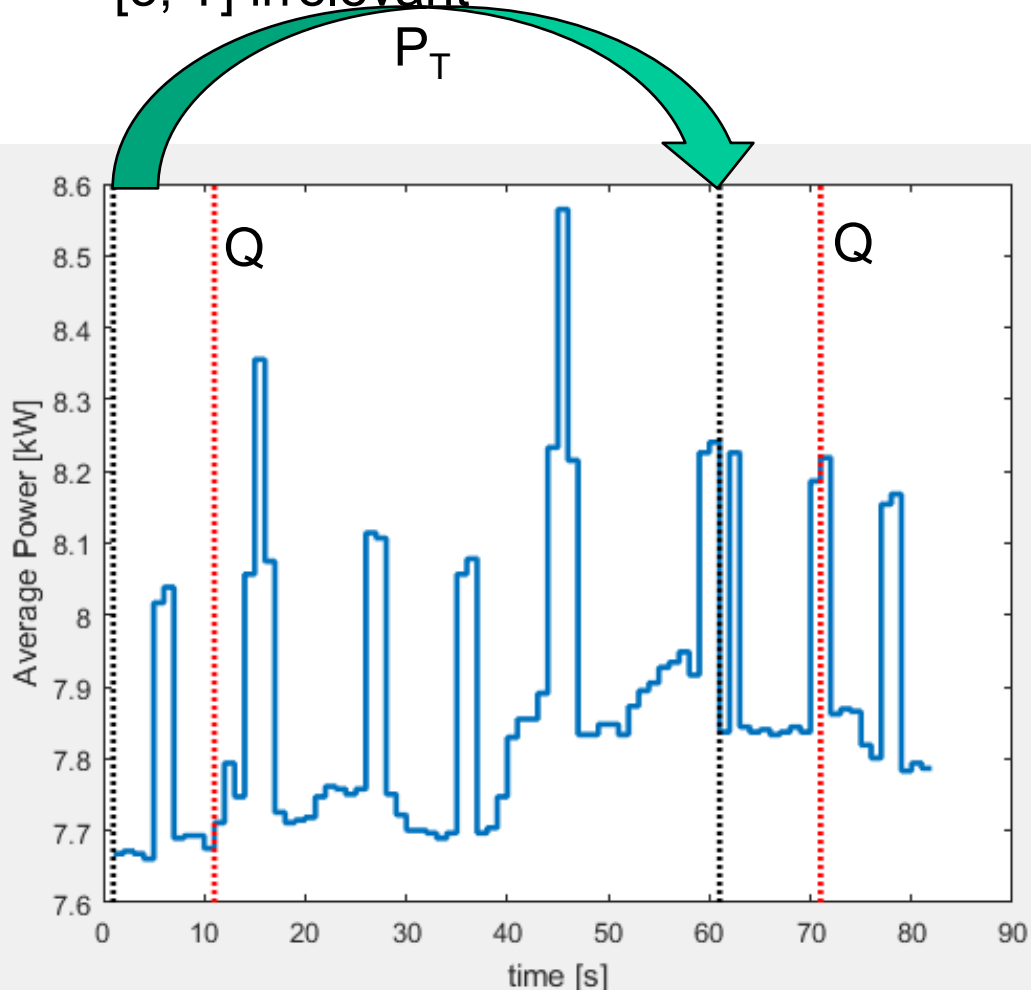
- 4) Distribute the total state leaving rate over the outgoing transitions

$$\text{lambda}_{ij} = \text{Lambda}_i * P_{ij}$$

Markov Model

Extension (for assessment)

- Correlation over time-scale of T
- Detailed behavior of measurand in $[\delta, T]$ irrelevant



MMP Gap-T model:

- 1) Use Q to describe behavior in $[0, \delta]$ and $[T, T + \delta]$
- 2) Use P_T to describe state transitions from 0 to T

Capturing the impact of interval misalignment - metrics



- True physical value $m(t)$: stochastic process with expected value μ
- Measurement over interval of duration T

$$\hat{m}(T, \delta) := 1/T \int_{\delta}^{T+\delta} m(t) dt.$$

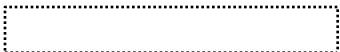
- Absolute measurement deviation
(random variable) for clock deviation δ

$$\epsilon(T, \delta) := \hat{m}(T, 0) - \hat{m}(T, \delta),$$

- Normalized time alignment error
- Additive normalized time alignment error

$$\kappa(T, \delta) := \frac{\text{std}(\epsilon(T, \delta))}{\mu}.$$

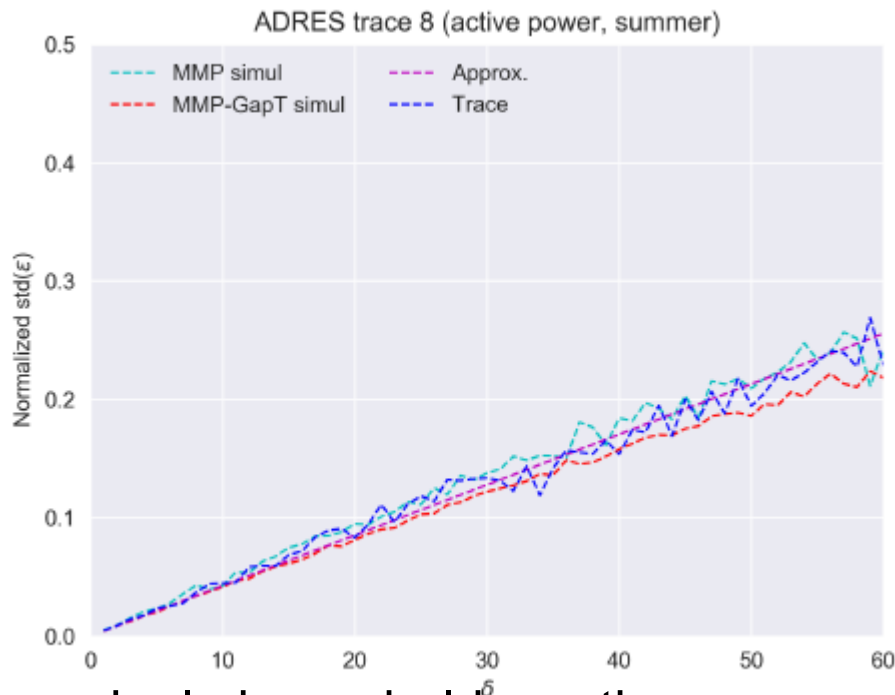
$$\alpha(T, \delta) := \frac{d}{d\delta} \kappa(T, \delta) = \frac{1}{\mu} \cdot \frac{d}{d\delta} \text{std}(\epsilon(T, \delta))$$



Behavior of Normalized Time Alignment

Error for increasing clock deviation

- Power at customer connection (household)
 - $T = 900\text{s} = 15\text{min}$



single household – active power
 $\alpha \approx 0.4\% / \text{s}$



different household
reactive power
 $\alpha \approx 2.6\% / \text{s}$

Data from ADRES project, see:

https://www.ea.tuwien.ac.at/fileadmin/t/ea/projekte/ADRES_Concept/Daten/TERMS_OF_USE_ADRES_DATA_120305.pdf



Outlook: How much do clock offsets matter – use-case grid loss calculation

- Depends on the application (use of the measurements):
 - Example loss calculation via
 - (1) Total Loss – computed from differences of Energy inflow and outflow:

$$L_1 = T \cdot \left(P_{imp}^{(0)} - P_{exp}^{(0)} + \sum_{j=1}^K \left(P_{inj}^{(j)} - P_{cons}^{(j)} \right) \right),$$

- (2) Technical Loss – computed from line currents and line resistances:

$$L_2 = T \cdot \sum_{j=1}^C I_j^2 R_j.$$



~~Outlook: How much do clocks offsets in measurement data matter for energy loss calculation?~~

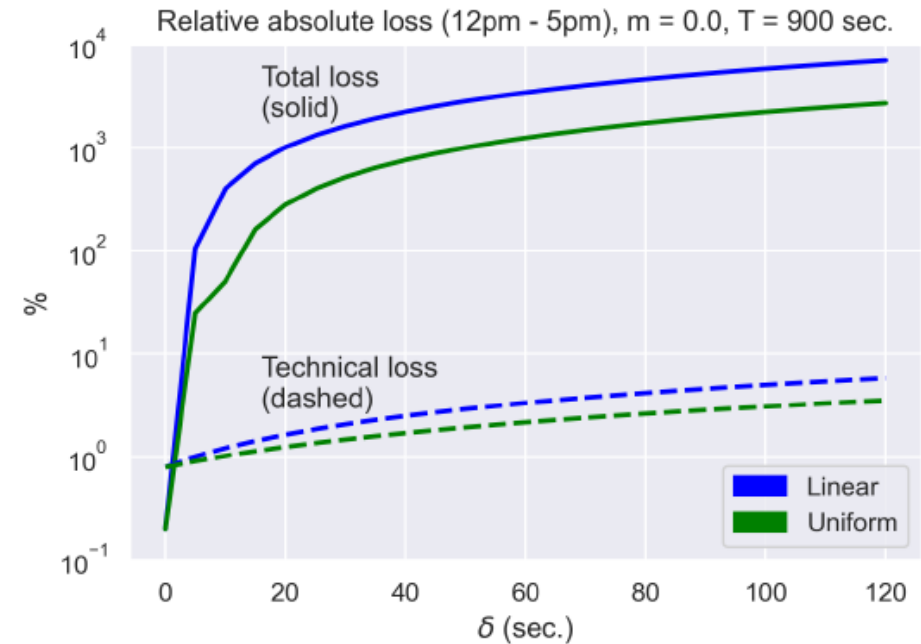
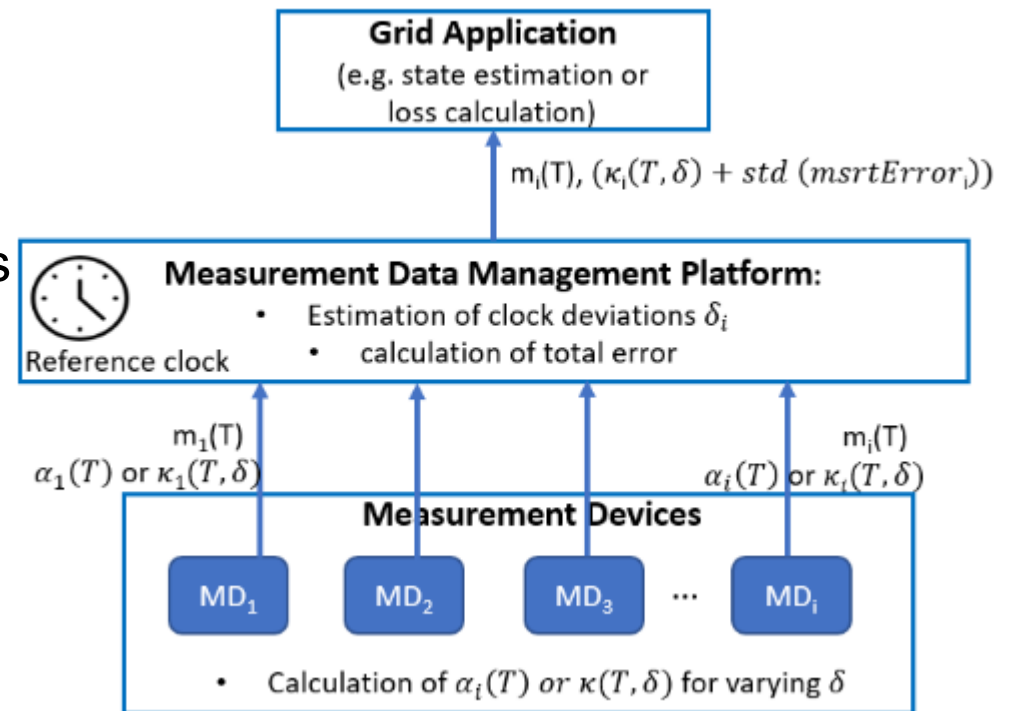


Figure 5: Relative loss deviation as a function of time alignment parameter δ , afternoon trace.

- Online use of time alignment error calculations (and other artefacts) in data management and data analytics solutions
- Applications to other IoT scenarios with distributed measurements
- Analyze trade-off: Smaller averaging period → reduced 'averaging' error but increased time alignment error
- Applications to control systems (in electricity grids and elsewhere)



Summary

1. Introduction
2. **Clock Synchronisation protocols**
 - Basic methods for synchronization and properties
 - Network Time Protocol (ntp)
3. **System-of-systems use-case: digital twin of electricity distribution grids**
 - Background MV and LV grids
 - Measurements
 - Digital twin
4. **Impact of clock offset on measurements**
 - Offset detection in distributed voltage measurements
Use-case loss calculation
 - definition of time alignment error
 - Modelling of consumption measurements via Markov modulated processes
5. **Summary**

Acknowledgment

Leveraging Networked Data for the Digital Electricity Grid - Net2DG

www.net2dg.eu





Exercises: impact of time stamps on voltage measurements

On Moodle, you find two sets of measurements of voltages in a grid (actually taken at 'neighboring' points in the grid).:

1. Start with the 1st measurement UL1 trace: Calculate the mean and standard deviation of the trace and plot the trace. Are these RMS values? What voltage level LV or MV? Discuss whether the trace shows any remarkable features that may point to needs for data preparation/cleaning.
2. Write some code to aggregate measurements to 15min averages. Plot the result for different offsets of the intervals in time: 0s, 5s, 30s, 60s for the first trace. Calculate the Root mean square error of the shifted aggregation in reference to the 0s shift for that trace.
3. Now look at the second data set: plot the two time series in one graph and discuss whether you visually see any indications of clock offsets between the two measurement points. Optional: Validate your suspicion by a cross-correlation calculation of the two time series and show that $|r(k)|$ has a maximum around the suspected offset.
4. Write some Matlab code to derive the Q and E matrix for a Markov modulated process (hint, you may want to start with the P matrix for the embedded discrete chain at transition moments). Use N=10 states and equidistant voltages covering the range of the data. Execute the Matlab code on Trace 1.

References – Part 1 – clock sync

- Verissimo, Rodrigues: 'Distributed Systems for System Architects', Kluwer, 2001, Sects 2.5 – 2.8
- Taha, Thorsager, Krøyer, Madsen, Schwefel: Network-Independent Clock Offset Estimation Using Correlated Sensor Data, Proceedings of international conferences on wireless personal multi-media communication (WPMC), Herning, November 2022.

Network Time Protocol (NTP)

- NTPv4, RFC5905, www.ietf.org



1. Nainar, K, Ciontea, C-I, Shahid, K, Iov, F, Olsen, RL, Schäler, C & Schwefel, H-PC 2021, 'Experimental Validation and Deployment of Observability Applications for Monitoring of Low-voltage Distribution Grids', *Sensors*, bind 21, nr. 17, 5770.
<https://doi.org/10.3390/s21175770>, see <https://vbn.aau.dk/da/publications/experimental-validation-and-deployment-of-observability-applicati>

Optional reading:

1. Schwefel, H-P, Antonios, I & Lipsky, L 2021, 'On the calculation of time alignment errors in data management platforms for distribution grid data', *Sensors (Switzerland)*. See <https://vbn.aau.dk/da/publications/on-the-calculation-of-time-alignment-errors-in-data-management-pl>
2. Antonios, I., Schwefel H-P, 'Assessing the impact of measurement artifacts on energy loss calculation in electrical distribution grids', to appear in ACM E-Energy 2023.

